Integ. Programming 2 - Industry, Innovation, & Infrastructure

3C ITPE 104 | Final Project - SDG (Sustainable Development Goals)

Project Team Members

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Focus Topic:

Sustainable Development Goal (SDG) 9 – Industry, Innovation, and Infrastructure

Background

SDG 9 underscores the importance of building resilient infrastructure, promoting sustainable and inclusive industrialization, and fostering innovation as critical drivers of economic growth and human well-being. This goal prioritizes quality, reliable, sustainable infrastructure that supports economic development across regions.

The key targets of SDG 9 include:

- Developing resilient and sustainable infrastructure
- Encouraging inclusive and sustainable industrial growth
- Increasing access to modern information and communication technologies (ICT)
- Supporting domestic technology development and industrial diversity
- Enhancing scientific research and upgrading technological capabilities

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Project Titles Overview and Documentation

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1. Web-Based Shipment Management System for Optimized Logistics

Problem Statement:
The logistics sector faces operational inefficiencies due to lack of real-time tracking, poor route planning, and limited stakeholder communication. These challenges increase costs, cause delays, and reduce overall customer satisfaction.

Scope Includes:

- Web platform accessible on multiple devices
- Real-time monitoring of shipments
- Route optimization to ensure timely deliveries
- Integration possibilities with existing logistics software
- Role-based user management
- Analytical dashboard for performance insights

Limitations:

- Dependent on stable internet connectivity and accurate GPS data
- Initially focused on road transport; expansion for other modes planned
- Does not cover warehouse automation

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2. Web-Based Complaint Tracking and Management System with Decision Support

Problem Statement:
Many organizations rely on manual or outdated complaint handling systems, leading to slow responses and poor service recovery. There is also a lack of systematic data analysis for complaints to improve decision-making.

Scope Includes:

- Centralized online complaint registration and tracking
- Automated complaint sorting and prioritization
- Real-time updates for complainants

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- Integration with CRM systems
- Decision support tools for managerial reporting
- Multi-channel access for submitting complaints (web, mobile, email)

Limitations:

- Staff training required for smooth adoption
- Complex data migration from legacy systems
- Currently supports only text-based complaints
- No automated resolution for complex issues yet

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3. Web-Based Idea Submission and Innovation Incubation System

Problem Statement:
Organizations often lack transparent, accessible platforms to gather and develop innovative ideas, affecting morale and innovation potential.

Scope Includes:

- Secure, user-friendly web portal for idea entry
- Transparent, community-based evaluation and voting
- Project tracking for incubated ideas
- Collaboration tools for development teams
- Integration with project/resource management systems
- Metrics and analytics on innovation performance
- Gamification features to boost participation

Limitations:

- Requires cultural shift to encourage idea sharing
- Legal considerations for intellectual property protection
- Limited to digital submissions, which may exclude some contributors
- Effective evaluation needs trained reviewers
- Success hinges on organizational commitment to innovation

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Technical Implementation

Technologies Utilized

- **Frontend:** HTML, CSS, JavaScript with modern frameworks (e.g., React, Vue)
- **Backend:** Server-side programming for API and business logic
- **Database:** MySQL (with XAMPP for local development)
- **Real-time Features:** Technologies like WebSockets for live updates
- **Integration:** External APIs to enhance system capabilities

Database Design

- Dedicated schemas per application for clarity and modularity
- Comprehensive data models supporting functionality and integrity
- SQL scripts available for easy setup and deployment

Architecture Highlights

- Modular, scalable design to allow independent development and future expansion
- Responsive interfaces for usability across devices
- Robust security measures for data protection and privacy

Development Status and Roadmap

Completed:

- Basic web interfaces and navigation
- Core backend and database setup
- Initial API integrations

Next Steps:

- Implementing real-time operational features
- UI enhancements with an emphasis on user experience

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- Development of advanced features such as route optimization

Future Plans:

- Mobile app for broader accessibility
- Advanced analytics and reporting modules
- Integration of predictive analytics through machine learning
- Multi-language support for global reach
- Cloud hosting options for scalability

Conclusion

The proposed projects align closely with SDG 9 by addressing industry challenges through innovative technological solutions. Each project promotes sustainable industry growth by enhancing operational efficiency, improving stakeholder experiences, and fostering innovation. This approach demonstrates how technology can be leveraged effectively to build resilient infrastructure and empower sustainable industrial development.

The projects are designed with a user-centered approach, ensuring practical deployment and scalability while supporting sustainable development principles.

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